

A PROJECT TITLE

ON

Loan Approval Prediction Using Ensemble

(RF, XGBoost, LightGBM)

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

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DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY

Name of Title: Loan Approval Prediction Using Ensemble Learning (RF, XGBoost, LightGBM)

1. ABSTRACT:

Loan approval is one of the most critical processes in the banking and financial sector, where accurate decisions protect both lenders and borrowers. Traditional evaluation relies on manual assessment of applicant information such as income, employment status, education, credit history, loan amount, dependents, and repayment capacity. This approach can be time-consuming, inconsistent, and influenced by subjective judgment. Machine learning provides an efficient alternative by identifying hidden patterns within historical data and generating consistent predictions.

This project presents a Loan Approval Prediction System using three ensemble learning algorithms: Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Light Gradient Boosting Machine (LightGBM). The system begins with data preprocessing, including handling missing values, encoding categorical variables, feature selection, normalization where required, and splitting the dataset into training and testing sets. Each model is trained and optimized using appropriate hyperparameter tuning techniques to improve predictive performance and generalization.

The trained models classify loan applications as approved or rejected based on applicant characteristics. Their performance is evaluated using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and the Confusion Matrix to provide a comprehensive comparison. Feature importance analysis is also performed to identify the variables contributing most to prediction outcomes, improving model interpretability and supporting transparent decision-making.

Experimental results demonstrate that ensemble learning methods outperform individual decision tree models by reducing overfitting and increasing prediction reliability. Among the evaluated models, the best-performing algorithm delivers high classification accuracy while maintaining robustness and computational efficiency. The proposed system can assist banks and financial institutions by automating loan screening, reducing processing time, minimizing default risk, and supporting fair, consistent, and data-driven lending decisions. Overall, this project demonstrates the practical value of ensemble learning for binary classification and highlights its potential for developing reliable, scalable, and intelligent decision support systems in modern banking environments while ensuring greater customer trust and long-term sustainability.

2. INTRODUCTION:

The banking and financial sector requires accurate loan approval decisions to minimize financial risk while ensuring that eligible applicants receive timely assistance. Traditional manual evaluation based

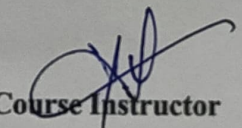
on income, employment status, credit history, education, loan amount, and repayment capacity can be time-consuming, inconsistent, and prone to human error. This project, **“Loan Approval Prediction Using Ensemble Learning (Random Forest, XGBoost, and LightGBM),”** aims to develop a machine learning system capable of predicting whether a loan application should be approved or rejected based on applicant information.

The project involves data preprocessing, feature engineering, model training, hyperparameter tuning, and performance evaluation. Three ensemble algorithms—Random Forest, XGBoost, and LightGBM—are compared using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix. Feature importance analysis is also performed to identify the factors that significantly influence loan approval decisions. The proposed system demonstrates how ensemble learning can automate loan screening, reduce processing time, minimize financial risk, and support fair, consistent, reliable, and data-driven decision-making, thereby improving the efficiency of modern banking operations.

3. SOFTWARE REQUIREMENTS:

Category	Details
Operating System	Windows 10/11, Linux, or macOS
Programming Language	Python 3.10+
IDE	Visual Studio Code / Jupyter Notebook
Libraries	Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, Matplotlib, Seaborn
Dataset Format	CSV (.csv)
Version Control	Git & GitHub (Optional)


Signature of the Guide


Course Instructor